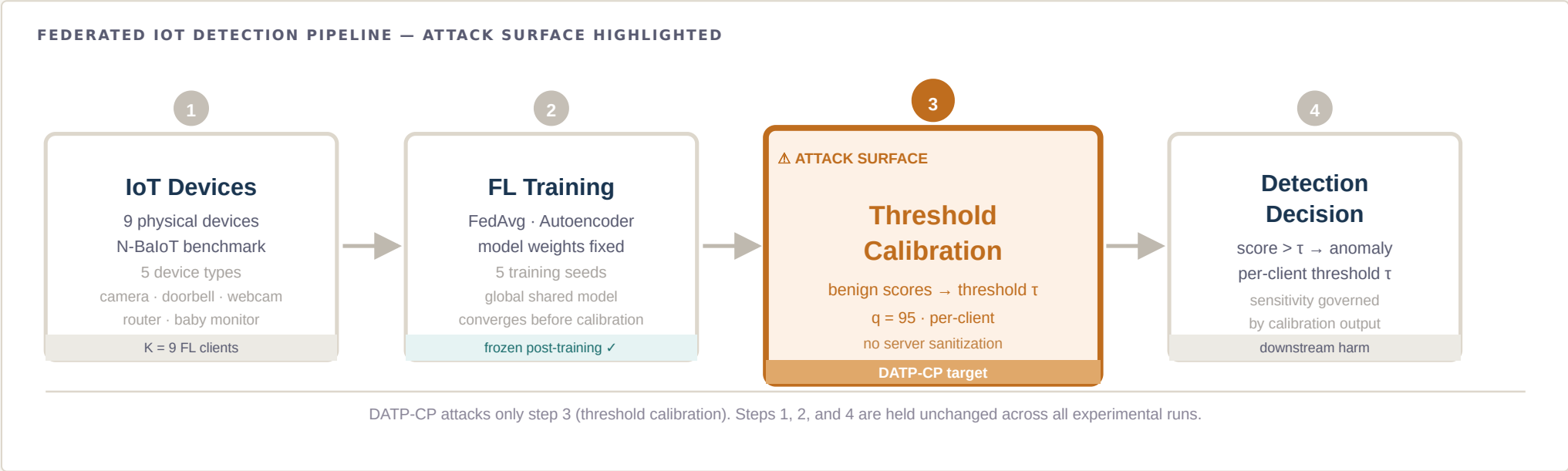


DATP-CP: Calibration-Channel Poisoning in Federated IoT Anomaly Detection

A policy-differentiated vulnerability analysis of the post-training threshold-calibration stage



ATTACK SURFACE

Post-training calibration channel

Benign calibration scores that determine the detection threshold τ . Entirely separate from training, aggregation, and model weights.

MECHANISM

Score-level contamination

Replace fraction f of a victim's calibration buffer with tail-sampled scores. Raising: 100% gate pass, -2.4 to -7.9 pp victim TPR. Lowering: $\Delta\text{CV}(\text{FPR}) +0.05-0.06$.

SCOPE

Policy-differentiated vulnerability

Global, Local, and Cluster threshold policies respond differently. Policy choice governs victim sensitivity, blast radius, and spillover.

Salaheddine Naslouby

LAMSAD Laboratory, ENSA Hassan First University, Berrechid · M. Lachgar (L2IS · UCA Marrakech) · H. Hrimech (LAMSAD)

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DATP-CP Branch

Calibration-Channel Poisoning

Post-Training Attack Surface

FOUNDATION

DATP

Established threshold personalization as the decisive calibration variable for non-IID federated IoT detection.

SECURITY QUESTION

What if calibration is contaminated?

If thresholds are computed from benign calibration scores, replacing those scores shifts the threshold without touching the model.

METHOD

Controlled post-training poisoning

Training, aggregation, model parameters, test scores, and test labels remain untouched. Only benign calibration scores are contaminated.

OUTPUT

Policy-differentiated vulnerability analysis

How Global, Local, and Cluster threshold policies react differently to the same calibration contamination.

The core argument

Federated IoT anomaly detectors must set a detection threshold for each client. DATP showed that *how* this threshold is set — globally, locally, or by cluster — governs per-client false-alarm burden. That makes threshold calibration **important for performance**.

DATP-CP asks the next logical question: if threshold calibration is important, is it also **vulnerable**? Can an adversary shift the threshold of a federated detector without touching training, aggregation, or the model itself — simply by contaminating the benign calibration buffer of one client?

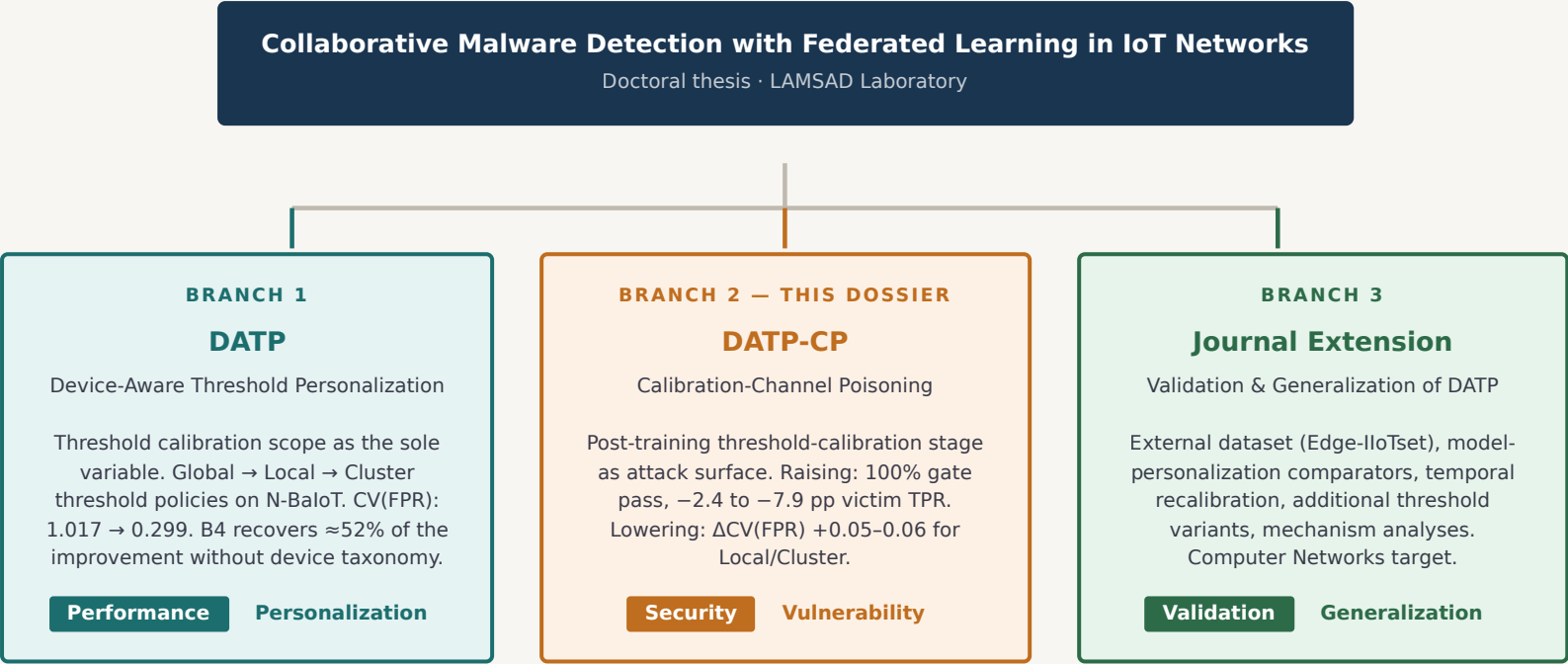
The answer is studied under a controlled score-level contamination model. The result is a **bounded vulnerability characterization**, not a claim about raw-traffic attack deployment.

What DATP-CP is and is not

DATP-CP is	DATP-CP is not
Post-training calibration-channel poisoning	Training-data poisoning
Threshold-stage isolation study	Model or aggregation poisoning
Policy-differentiated vulnerability analysis	Universal FL robustness claim
Score-level contamination mechanism	Raw-traffic attack deployment
Continuation of the DATP research line	Replacement for the journal extension

Q Why is this worth studying?

A Because threshold calibration determines the operating point of the detector. If thresholds matter for performance and fairness, they also matter for security. DATP-CP isolates exactly that stage.



Key principle: The three branches share a thesis line but have **separate scientific roles**. DATP-CP is the adversarial robustness branch focused specifically on the threshold-calibration stage. It does not absorb the journal extension's validation goals and does not duplicate DATP's personalization study.

Dimension	DATP (Branch 1)	DATP-CP (Branch 2)	Journal Extension (Branch 3)
Primary variable	Threshold policy (Global vs. Local vs. Cluster)	Calibration-buffer contamination fraction f	Dataset, comparator, threshold variant
Adversary	None — performance study	Gray-box score access, single client	None — validation study
Primary metric	CV(FPR) across eligible clients	$\Delta\tau$, ΔTPR , $\Delta\text{CV(FPR)}$ paired per seed	External replication of CV(FPR) reduction
Claim type	Calibration scope governs detection equity	Calibration channel is a viable attack surface	DATP effect survives generalization pressure
What it does not claim	Adversarial robustness of calibration	Training / model / aggregation poisoning	Adversarial branch (deferred to DATP-CP)

DATP in one paragraph

DATP (*Device-Aware Threshold Personalization*) is a controlled threshold-calibration study for non-IID federated IoT anomaly detection. It fixes the shared FedAvg autoencoder and varies only the threshold policy — the sole experimental variable — to isolate the effect of calibration scope on per-client false-alarm burden.

Using N-BalIoT with 9 physical-device clients, DATP showed that a shared global threshold imposes severely unequal FPR across devices (CV(FPR) = 1.017), that per-client calibration reduces this disparity to CV(FPR) = 0.299, and that an unsupervised cluster policy recovers ≈52% of the improvement without requiring device taxonomy labels.

DATP's controlled design means its results are causally attributable to threshold calibration scope, not to training or architecture differences.

DATP key results

Metric	Global (B1)	Per-client (B2)	Cluster (B4)
CV(FPR) N-BalIoT	1.017	0.299	~0.64 recov.
Bootstrap 95% CI on Δ	[0.647, 0.769] — all 5 seeds positive		
CICIoT2023 (file-level)	Near-null — applicability boundary		
Seeds	5 training seeds		

Q What did DATP leave open?

A DATP established threshold calibration as consequential for detection equity. It did not study whether that calibration stage can be adversarially manipulated — that is precisely DATP-CP's contribution.

DATP question-to-contribution map (and the open question that motivates DATP-CP)

DATP question	Mechanism studied	Why it mattered	What it left open
How large is per-client FPR disparity under a shared threshold?	Global threshold = mean of local p95 thresholds	Quantified the fairness cost of one-size-fits-all thresholds	No attack model studied
How much does per-client calibration reduce disparity?	Local threshold = each client's own p95	Showed per-device calibration is the decisive variable	Calibration seen as benign-only
Can unsupervised clustering approximate per-client performance?	Cluster threshold on calibration-error fingerprints	Taxonomy-free compromise between global and local	Cluster propagation under adversarial input not studied
OPEN QUESTION THAT LEADS TO DATP-CP ↓			
If threshold calibration is important, can it become an attack surface?	Benign calibration scores → threshold τ	Calibration is autonomously curated post-training; no server-side sanitization exists	→ DATP-CP answers this



Five specific added-value claims

- Security perspective on calibration.** DATP-CP converts threshold calibration from a performance mechanism into a security question — a perspective DATP does not take.
- Isolation of a post-training attack surface.** Training-phase and aggregation-phase defenses do not observe the calibration stage. DATP-CP is the first study to isolate this gap in federated IoT threshold-based anomaly detection.
- Policy-differentiated behavior under the same attack.** The same contaminated calibration buffer affects Global, Local, and Cluster thresholds differently. DATP-CP measures victim sensitivity, blast radius, spillover, and cluster propagation.
- Causal cleanliness.** By freezing training, aggregation, model parameters, and test data, DATP-CP attributes any threshold shift solely to calibration contamination — a stronger causal claim than observational comparisons can support.
- Bounded, honest characterization.** The study produces a pre-registered vulnerability claim under score-level contamination — not an overbroad security statement.

Contribution positioning against adjacent work

Prior work type	Attack surface	Distinction from DATP-CP
Training-data poisoning	Training set / labels	DATP-CP leaves training data and model unchanged
FL model poisoning	Client gradients	Needs no gradient access; acts after convergence
Byzantine aggregation	Aggregation inputs	Does not touch aggregation; harm is post-aggregation
Centralized AD poisoning	Central training stream	No trusted sanitizer exists in FL calibration
Threshold personalization (DATP)	Threshold policy	Studies vulnerability, not utility
DATP-CP	Calibration scores (post-training)	Score-level, post-training, single-client, policy-differentiated

Q Is this just another poisoning paper?

A No. The study deliberately excludes training, model, and aggregation poisoning. It attacks only the post-training calibration channel — a stage that existing FL-IDS defenses do not observe or sanitize. The policy-differentiated analysis (Local vs Cluster vs Global) is the scientific point, not the attack mechanism alone.

The Post-Training Coverage Gap

Why calibration is under-protected

Federated learning security research overwhelmingly focuses on the **training phase**: Byzantine-robust aggregation, gradient sanitization, and model-integrity verification. These defenses operate before the model is deployed and complete before calibration begins.

The threshold-calibration stage is structurally different:

- It occurs **after training convergence** — Byzantine defenses have already exited.
- Each client **autonomously collects** its benign calibration window — no central validation gate.
- Calibration data **never leaves the client** — the server cannot sanitize it.
- The threshold is computed locally from that benign window — a single compromised client can shift its own threshold, or through Global/Cluster policies, propagate effects to non-victims.

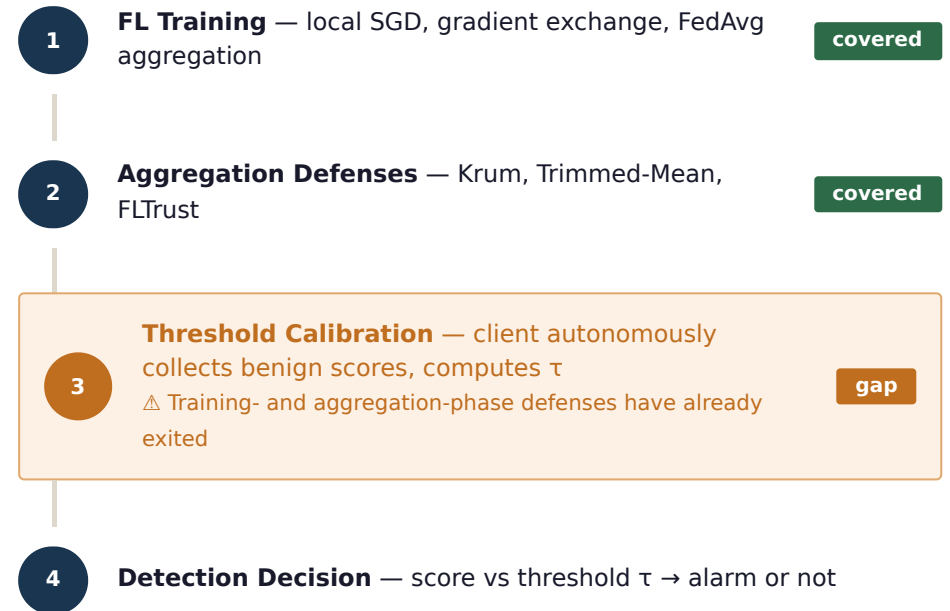
THE STRUCTURAL GAP

If a client's benign calibration buffer is contaminated with scores drawn from the same client's benign distribution but biased toward the high or low tail, the threshold shifts — even though the model, gradients, aggregation, and test data are all unchanged. No existing FL training- or aggregation-phase defense covers this stage.

Q Why not study model poisoning instead?

A Model poisoning changes the learned detector. DATP-CP freezes the detector and studies only the post-training threshold-calibration channel — a different attack surface that training-phase defenses do not observe.

Where calibration sits in the full pipeline



DATP-CP'S ROLE

DATP-CP focuses entirely on step 3 — the calibration gap. Steps 1, 2, and 4 are held constant. This clean isolation is what makes the causal attribution valid.

MAIN RESEARCH QUESTION

Does poisoning only the benign threshold-calibration set of eligible clients shift threshold policies enough to induce interpretable downstream failures, and do **GLOBAL_THRESHOLD** **LOCAL_THRESHOLD** **CLUSTER_THRESHOLD** exhibit different vulnerability profiles?

Confirmatory subquestions

ID	Research Question	Measured via
RQ1	Does calibration-only poisoning produce material, correctly signed $\Delta\tau$ under the gray-box score-access threat model?	$\Delta\tau$ materiality gate, seed-sign count
RQ2	Do Global, Local, and Cluster threshold policies differ in victim sensitivity, blast radius, spillover, and operating-point disparity?	Policy-stratified $\Delta\tau$, BlastRadius, SpilloverCount
RQ3	Do threshold-raising and threshold-lowering attacks produce separable failure modes?	Δ TPR (raising) vs Δ CV(FPR) (lowering)
RQ4	Across all eligible N-BaloT victims, does single-client compromise produce consistent effects under paired seeds?	Victim-majority rate across 10 seeds

CONFIRMATORY SCOPE

All four RQs are answered by the NBAIoT_MAIN stage. RQ1 drives the mechanism claim (Gate 1). RQ2 drives the policy-differentiation claim (Gate 2). RQ3 maps failure modes. RQ4 confirms victim consistency.

Optional subquestions

RQ5 — Limited multi-client compromise

Do pre-registered pairs and triples amplify or change the policy-differentiated vulnerability profile?
Optional full extension — not required for main single-client claim.

RQ6 — Lightweight mitigation

Can TRIMMED_CALIBRATION reduce the declared primary harm metric without material clean-setting regression?
Optional — appears in main results only if it meets pre-specified recovery and regression gates.

Pre-specified claim gates

Gate	Pass condition
Gate 1	Seed-sign count $\geq$ 8/10 AND victim-majority count $\geq$ 8/10 for material correctly-signed $\Delta\tau$
Gate 2	Directional excess over matched RANDOM_BENIGN control at $\geq$ 8/10 seeds
Gate 3	At least one downstream harm metric has expected-sign support at $\geq$ 8/10 seeds

Claim class depends on gate outcome: Full vulnerability (all 3), Mechanism-only (Gate 1 only), Calibration-instability (Gate 2 fails), or Null/conditional.

Attacker capability table

✓ Attacker CAN	✗ Attacker CANNOT
✓ Influence which benign records enter the victim's calibration buffer	✗ Alter training data or training labels
✓ Score local benign candidates using the locally available model	✗ Alter model weights or gradients
✓ Replace up to fraction f of calibration buffer entries	✗ Alter server aggregation
✓ Select high-score or low-score tail candidates (gray-box access)	✗ Alter test scores or test labels
✓ Act after training convergence, before threshold computation	✗ Observe or use other clients' raw data
	✗ Insert attack-labeled samples into calibration
	✗ Poison the evaluation set

Q Why is the attacker allowed to see local scores?

A Because a compromised FL client holds the trained detector and can score its own local benign data. Gray-box score access is native to the client role in federated anomaly detection. The attacker needs no access to gradients, server state, or other clients' data.

Knowledge model

MAIN STUDY: GRAY_BOX_SCORE_ACCESS

The attacker has local reconstruction scores for local benign candidate records. Justified because each FL client holds the trained anomaly detector and can score its own local candidates. This is the only knowledge model that supports main claims.

Attack objectives

THRESHOLD_RAISE

Insert HIGH_SCORE_BENIGN tail samples $\rightarrow \Delta\tau > 0$
Expected effect: victim TPR decreases (missed detections)

THRESHOLD_LOWER

Insert LOW_SCORE_BENIGN tail samples $\rightarrow \Delta\tau < 0$
Expected effect: $\Delta\text{CV}(\text{FPR}) > 0$, alarm burden increases

Threat-model variants

Variant	Role	Claim status
MAIN_GRAY_BOX	Victim-local score access and benign reservoirs	Main claims only
MINIMAL_NEGATIVE_CONTROL	RANDOM_BENIGN — near-null expected	Negative control
DIAGNOSTIC_UPPER_BOUND	Direct score edits or stronger access	Appendix only

GRAY-BOX ACCESS IS THE BINDING CONSTRAINT

MAIN_GRAY_BOX is the only variant that supports main claims. A compromised FL client can reconstruct scores locally — no gradient access, no server state, no cross-client data. MINIMAL_NEGATIVE_CONTROL is run paired with every directional attack for sign-consistency. DIAGNOSTIC_UPPER_BOUND (direct score edit) sets an approximate performance ceiling and appears only in the appendix, never in confirmatory inference.

Injection rule: REPLACE_FIXED_BUDGET

For victim i and poison fraction f , the injector selects $m_i = \max(1, \text{round}(f \times n_i))$ positions in the victim's benign calibration array and replaces them with values sampled with replacement from the declared victim-local reservoir. Calibration size remains constant — no records are added or removed, only replaced.

Source strategies

Strategy	Reservoir	Objective	Expected $\Delta\tau$
HIGH_SCORE_BENIGN	Upper 10% of victim's benign score distribution	THRESHOLD_RAISE	$\Delta\tau > 0$ (raises)
LOW_SCORE_BENIGN	Lower 10% of victim's benign score distribution	THRESHOLD_LOWER	$\Delta\tau < 0$ (lowers)
RANDOM_BENIGN	Uniformly sampled victim-local benign scores	Negative control	≈ 0 (near-null)

Q Why is RANDOM_BENIGN included?

A It audits natural calibration instability. If RANDOM_BENIGN also shifts thresholds materially, the directional attack result is confounded by intrinsic threshold variability — not by targeted poisoning. Each directional attack has a paired RANDOM_BENIGN negative control.

CALIBRATION SIZE PRESERVED

m_i entries replaced, none added or removed. Buffer size n_i constant across clean and poisoned branches.

VICTIM-LOCAL RESERVOIR ONLY

All replacement samples drawn from the victim's own benign scores. No cross-client access, gradients, or server state required.

AUROC INVARIANCE EXPECTED

Only calibration scores are touched; test scores/labels are frozen. AUROC stays unchanged, confirming calibration-channel isolation.

Conceptual distribution shift (score-level illustration)



Conceptual illustration only. No numeric values fabricated. Threshold τ = p95 of the calibration distribution.

SCORE-LEVEL PROXY LIMITATION

DATP-CP operates at the reconstruction-score level. This isolates the calibration stage but does not prove that an attacker can generate raw network traffic with arbitrary target scores. **All raw-traffic realizability claims are out of scope.** With-replacement resampling at high poison fractions can duplicate score values — this is disclosed as a proxy limitation.

Why N-BaloT?

N-BaloT provides labeled benign and Mirai/BASHLITE traffic for nine physical IoT devices spanning five device types. Each physical device functions as one FL client — this is not a synthetic or file-defined partition but a natural device-level split motivated by hardware identity and traffic distribution.

Property	N-BaloT value
Clients (K)	9 physical devices
Client definition	One physical device = one client
Traffic families	Mirai, BASHLITE (Gafgyt) botnet families
Device types	5 types (camera, doorbell, webcam, router, baby monitor)
Minimum calibration size	n_cal ≥ 100 for eligibility
Threshold percentile	q = 95 (locked)

Q Why N-BaloT specifically?

A Because it provides physical-device client identity — the heterogeneous per-device traffic distributions that make threshold personalization meaningful and make calibration-channel contamination consequential. Synthetic partitions would not preserve the device-level calibration behavior that DATP-CP studies.

CLAIM SCOPE

All claims are bounded to the nine-device N-BaloT setting with autoencoder detectors and FedAvg aggregation. No dataset-universal behavior is claimed.

Nine-device federation

Device 1 Camera type	Device 2 Camera type	Device 3 Camera type
Device 4 Webcam type	Device 5 Doorbell type	Device 6 Doorbell type
Device 7 Baby monitor	Device 8 Router type	Device 9 Router type

Client = physical device · K = 9 · Eligibility: n_cal ≥ 100

Data split semantics

Split	Role	Attack access
Benign training	Model learning (FedAvg)	Frozen — cannot be modified
Benign calibration	Threshold computation	⚠ Reservoir for DATP-CP attack
Test scores + labels	Evaluation (AUROC, TPR, FPR)	Frozen — cannot be modified

No overlap between calibration reservoirs and test data. No attack-labeled records in calibration.

GLOBAL_THRESHOLD

$$\tau_i^{local} = p95(S_i^{cal})$$
$$\tau^{global} = \text{mean}_i(\tau_i^{local})$$

Concept: One shared threshold for all clients, computed as the mean of eligible clients' local p95 calibration thresholds.

Every eligible client receives the same τ^{global} .

Under poisoning: Victim's contaminated local τ enters the global mean. Effect is diluted by 1/N but propagates to all clients (fleet-wide spillover). A single client poison has reduced but guaranteed fleet-wide reach.

1/N dilution

Fleet-wide propagation

LOCAL_THRESHOLD

$$\tau_i = p95(S_i^{cal})$$

Concept: Each eligible client computes its own threshold from its own calibration distribution. No cross-client averaging.

Strongest per-client personalization; zero structural dependency between clients' thresholds.

Under poisoning: Victim's contaminated calibration shifts only its own threshold. No cross-client spillover by construction. Direct victim sensitivity is maximum — threshold shift is fully absorbed by the victim.

Maximum victim sensitivity

No spillover

CLUSTER_THRESHOLD

$$v_i = [\mu, \sigma, \text{skew}, p95](S_i^{cal})$$
$$\text{K-means++ (K=3)} \rightarrow \text{cluster } c$$
$$\tau_i = \text{mean}(\tau_j^{local}) \text{ for } j \text{ in } c$$

Concept: Clients are fingerprinted by calibration-error statistics and grouped by K-means. Each client receives its cluster's mean local threshold.

Under poisoning: Victim's contaminated scores shift its fingerprint, potentially changing cluster assignment (churn). Co-cluster clients may receive a perturbed cluster threshold. Normalization-mediated effects also arise when the scaler refits on poisoned statistics.

Intra-cluster spillover

Churn + refit effects

Policy comparison summary under calibration poisoning

Policy	Victim sensitivity	Blast radius	Spillover mechanism	Empirical $\Delta\tau$ signal (raising)
GLOBAL	Diluted (1/N)	All eligible clients	Mean aggregation — direct	Smaller, fleet-wide
LOCAL	Maximum (undiluted)	Victim only	None by construction	Largest per-victim
CLUSTER	Moderate	Co-cluster clients	Cluster aggregation + churn + normalization	Intermediate; cluster-dependent

Q Why compare policies instead of only showing local threshold movement?

A The scientific point is policy differentiation: the same poisoned calibration channel may remain local, propagate globally, or spill over through clusters depending on the threshold policy. This comparison is the main contribution — not the magnitude of the attack in isolation.

Main matrix — NBAIOT_MAIN

Dimension	Values
Dataset	N-BaloT (9 physical devices)
Threshold policies	GLOBAL_THRESHOLD · LOCAL_THRESHOLD · CLUSTER_THRESHOLD
Scope	SINGLE_CLIENT — all eligible victims
Source × Objective pairs	HIGH_SCORE_BENIGN × THRESHOLD_RAISE LOW_SCORE_BENIGN × THRESHOLD_LOWER RANDOM_BENIGN × THRESHOLD_RAISE (control) RANDOM_BENIGN × THRESHOLD_LOWER (control)
Poison fractions (f)	0 · 0.10 · 0.20 · 0.40
Training seeds	0–9 (10 seeds)
Poisoning seeds	100–109 (paired per training seed)
Threshold percentile	q = 95 (locked, cannot change)
CV std convention	ddof = 0
Numerical epsilon	$\epsilon_{\text{num}} = 1\text{e-}12$

Objective-labeled row count: 10 seeds × 9 victims × 3 policies × 4 source-objective pairs × 4 fractions = 4,320 rows.
These are not independent. Confirmatory inference uses 10 paired seed-level aggregates.

Experiment stages

Stage	Purpose	Gate status
FINAL_AUDIT	Protocol consistency, clean-artifact provenance, cluster reproducibility, forbidden-scope check	Required before NBAIOT_MAIN
SYNTHETIC_SMOKE	Validate poisoning invariants on controlled arrays; verify direction, no mutation, AUROC invariance	Blocks main execution if fails
NBAIOT_MAIN	Primary confirmatory study — all eligible single-client victims, all valid source-objective pairs	Primary study
NBAIOT_FULL_OPTIONAL	Multi-client (pairs + triples), optional defense, optional f=0.05	Optional only
STRETCH_DIAGNOSTIC_ONLY	CICIoT2023 as pseudo-client contrast (diagnostic, no natural-device claims)	Optional, diagnostic

Invalid source-objective pairs (rejected by config)

Rejected — fail config validation:
HIGH_SCORE_BENIGN + THRESHOLD_LOWER · LOW_SCORE_BENIGN + THRESHOLD_RAISE
These combinations are physically inconsistent and cannot be executed as main cells.

Q Why only 10 seeds rather than more?
A 10 paired seeds provide a seed-level aggregate distribution suitable for bootstrap CIs and sign consistency rules. The two-layer statistical plan (victim-level paired deltas + seed-level aggregates) drives inference, not raw cell count.

Causal Isolation — Paired Clean vs. Poisoned Design

The paired design principle

Every poisoned run is paired with a clean counterpart that is **identical in every respect except the calibration buffer contents**. This pairing is what makes causal attribution valid: any difference in thresholds or downstream metrics is attributable to the calibration intervention, not to randomness in training, model initialization, or aggregation.

What is held constant across each pair

SHARED BY CLEAN & POISONED

- Training seed (0-9)
- Model parameters (weights)
- Aggregation process (FedAvg)
- Clean test scores
- Clean test labels
- Victim plan (identity)
- Threshold policy
- Source-objective pair
- Poison fraction f

ONLY DIFFERENCE

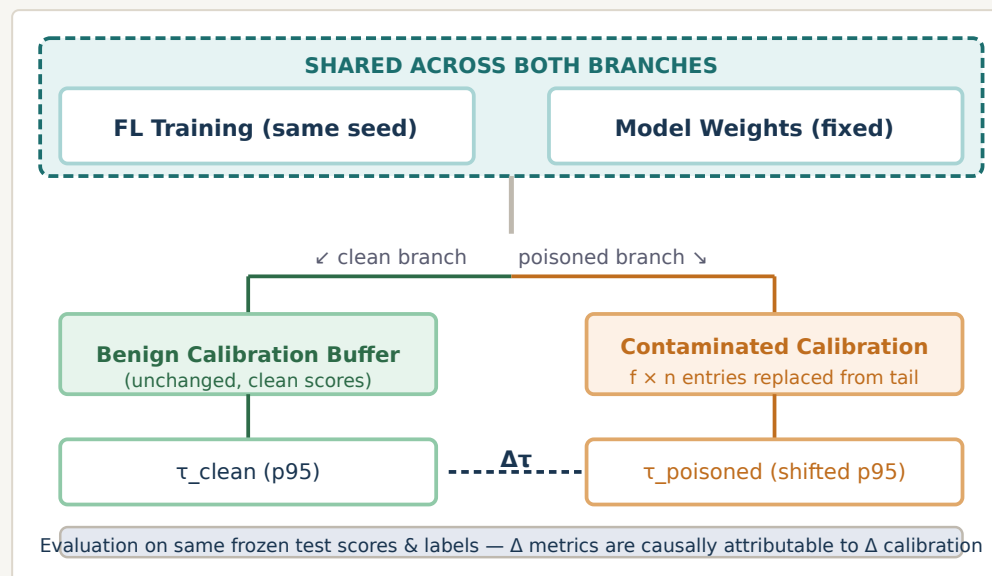
Calibration buffer contents

m_i entries in victim's benign calibration array are replaced with tail-sampled scores from the victim-local reservoir

Q How do we know observed effects come from calibration poisoning?

A Because training, aggregation, model parameters, test scores, and test labels are frozen across both branches of each pair. The calibration intervention is the isolated difference. Any threshold shift or downstream metric change is causally attributable to calibration contamination.

Clean vs. poisoned pipeline diagram



WHAT THE DESIGN PREVENTS

- Confounding from training randomness
- Confounding from model initialization
- Confounding from aggregation variance
- Test-set contamination

WHAT THE DESIGN ENABLES

- Causal attribution of $\Delta\tau$ to calibration
- Paired signed deltas per victim per seed
- Bootstrap CIs over seed-level aggregates
- AUROC invariance as a sanity check

WHY PAIRED DESIGN IS THE CORRECT UNIT OF ANALYSIS

Each DATP-CP result is reported as a signed delta ($\Delta\tau$, ΔTPR , $\Delta\text{CV}(\text{FPR})$) computed within a (victim, seed, fraction, source-objective) pair. Because training, model weights, test scores, and test labels are all identical within a pair, any non-zero delta is causally attributable to the calibration intervention alone. Sign-consistency across seeds and victim-majority counts across seeds are the primary inferential tools — the 4,320-row grid is not treated as 4,320 independent observations.

Metric interpretation table

Metric	Plain meaning	Why it matters	Research question it answers
$\Delta\tau$	Change in the victim's detection threshold (poisoned – clean)	Mechanism endpoint — did poisoning actually shift the threshold?	RQ1: Does calibration-only poisoning produce material $\Delta\tau$?
$ \Delta\tau \geq \delta\tau$	Threshold shift exceeds victim-specific materiality floor (10% of clean IQR)	Distinguishes operationally meaningful shifts from numerical noise	RQ1: Materiality gate — Gate 1
$\Delta\text{TPR (victim)}$	Change in victim true-positive rate (correctly detecting attacks)	Primary harm measure under threshold raising: missed detections	RQ3: THRESHOLD_RAISE failure mode
$\Delta\text{CV(FPR)}$	Change in coefficient of variation of false-positive rates across eligible clients	Alarm-burden disparity across the fleet; higher = more unfair distribution	RQ3: THRESHOLD_LOWER failure mode
$\Delta\text{IQR(FPR)}$	Change in interquartile range of per-client FPR	Absolute dispersion backup when CV is undefined or unstable	RQ3: Lowering dispersion evidence
BlastRadius	Number of non-victim eligible clients with material $ \Delta\tau $	Measures how far the attack propagates beyond the compromised client	RQ2: Policy-differentiated spillover
$\Delta\tau_{\text{churn}}$	Cluster-threshold component from client reassignment between clusters	Decomposes CLUSTER_THRESHOLD movement into aggregation vs assignment effects	RQ2: Cluster-specific mechanism
ΔAUROC	Change in area under ROC curve (sanity check)	Should be ≈ 0 — confirms test scores and labels were not changed	Sanity check only — not a contribution

AUROC INVARIANCE — IMPORTANT CLARIFICATION

AUROC invariance is a **sanity check**, not a contribution. It confirms that test scores and labels were not accidentally changed. If AUROC is invariant, the failure mode must be threshold-induced operating-point movement — which is exactly what DATP-CP measures. A reviewer who says "AUROC doesn't change, so nothing happened" is conflating the object of study (the threshold) with the scoring capacity (AUROC).

STATISTICAL DESIGN

Two-layer unit of analysis:

- **Layer 1:** Victim-level paired deltas — for sign consistency and majority support
- **Layer 2:** Seed-level aggregates (10 seeds) — primary inference unit

95% bootstrap CIs are computed over the 10 seed-level aggregates. The 4,320-row grid is not treated as independent evidence.

Expected Failure Modes

FAILURE MODE 1: THRESHOLD RAISING

Attack HIGH_SCORE_BENIGN inserted → calibration mean shifts up



Threshold $\Delta\tau > 0$ — threshold rises above clean level



Effect Attack traffic falls below raised threshold → **missed detections**



Metric $\Delta\text{TPR}_{\text{victim}} < 0 \cdot \Delta\text{BA}_{\text{victim}} < 0$

FAILURE MODE 2: THRESHOLD LOWERING

Attack LOW_SCORE_BENIGN inserted → calibration mean shifts down



Threshold $\Delta\tau < 0$ — threshold lowers (conditional on tail displacement)



Effect More benign traffic classified as anomalous → **false alarm increase**



Metric $\Delta\text{CV}(\text{FPR}) > 0 \cdot \Delta\text{IQR}(\text{FPR}) > 0$

Empirical result: Raising attack achieves 100% materiality gate pass rate at all non-zero fractions. Victim TPR degrades by 2.4–7.9 pp overall (4.9–7.9 pp at $f = 0.40$). Bootstrap CIs exclude zero. (Source: DATP-CP paper, 2026)

Empirical result: Lowering raises false-alarm indicators for Local and Cluster ($\Delta\text{CV}(\text{FPR}) = +0.05\text{--}0.06$ with Replace-Fixed-Budget). Global bounds lowering through 1/N dilution. Lowering is conditional — random fixed-budget insertion may not displace upper-tail support. (Source: DATP-CP paper, 2026)

Q Why does lowering sometimes increase TPR?

A A lower threshold may classify more points as malicious, which mechanically raises TPR. But this is not a security benefit if it simultaneously increases false alarms (FPR) or FPR disparity. DATP-CP evaluates lowering success via $\Delta\text{CV}(\text{FPR})$ and $\Delta\text{IQR}(\text{FPR})$, not via TPR alone.

Q Is lowering always successful?

A No — lowering is pre-registered as conditional. Random fixed-budget insertion of low-score benign samples may not displace the upper-tail support that sets the p95 threshold. The protocol distinguishes a robust calibration channel from a weak lowering instantiation using a diagnostic targeted-removal variant.

POLICY GOVERNS HARM MAGNITUDE AND PROPAGATION

Under raising, LOCAL_THRESHOLD delivers the largest per-victim TPR drop (maximum undiluted effect); GLOBAL_THRESHOLD spreads the impact fleet-wide at 1/N dilution; CLUSTER_THRESHOLD causes intra-cluster spillover. Under lowering, LOCAL and CLUSTER show material FPR disparity increases while GLOBAL is bounded by the same 1/N factor. The same poisoning fraction thus produces qualitatively different failure modes depending on which policy is in force — this policy differentiation is the central contribution of DATP-CP.

Dimension	DATP (Branch 1)	DATP-CP (Branch 2)	Journal Extension (Branch 3)
Primary question	How much does threshold calibration scope reduce per-client FPR disparity in non-IID federated IoT detection?	Can benign threshold-calibration data be contaminated after training to shift detection thresholds and harm detection?	Does DATP's threshold-scope result hold under a stronger protocol with external validation, comparators, and stress tests?
Scientific role	Foundation — Performance & personalization	Security — Vulnerability characterization	Validation — Generalization & stress-testing
Main object of study	Threshold policy (Global / Local / Cluster) as the sole experimental variable	Post-training threshold-calibration channel as an attack surface	DATP result robustness under additional datasets, comparators, and protocol variants
Attack model	None — no adversary modeled	Gray-box score-access, single compromised client, calibration-only contamination	None — adversarial robustness explicitly deferred to DATP-CP
Primary dataset	N-BalIoT (confirmatory) · CICIoT2023 (applicability boundary)	N-BalIoT (physical-device clients, confirmatory)	N-BalIoT + Edge-IIoTset (external) · CICIoT2023 (device-MAC redesign)
Threshold policies	Global (B1) · Local (B2) · Family-mean (B3) · Cluster (B4)	GLOBAL_THRESHOLD · LOCAL_THRESHOLD · CLUSTER_THRESHOLD	DATP baselines + conformal thresholding + shrinkage + q-sensitivity
Comparators	Centralized AE baseline (B0); no model-personalization comparators	No model-personalization comparators (out of scope)	FedProx · Ditto or FedRep-AE/FedPer-AE · Laridi-compatible variant
Main claim type	B2 reduces CV(FPR) by $\Delta=0.718$, 95% CI [0.647, 0.769], all seeds positive	Policy-differentiated vulnerability characterization under score-level calibration contamination	DATP threshold-scope effect survives external validation and stress tests
Key result	CV(FPR): 1.017 $\rightarrow$ 0.299. B4 recovers $\approx 52\%$.	Raising: 100% gate pass, -2.4 to -7.9 pp victim TPR. Lowering: $\Delta\text{CV}(\text{FPR}) +0.05-0.06$ (Local/Cluster).	10-seed extended replication + Edge-IIoTset + temporal MVE + 6 mechanism analyses
What it does not claim	No attack model · No privacy guarantee · No deployment readiness	No training/model/aggregation poisoning · No universal vulnerability · No deployment	Does not absorb adversarial or security scope (deferred to DATP-CP)

Separate branches, same thesis line. · Each branch has an exclusive scientific role. · No branch should absorb the goals of another.

What DATP-CP Intentionally Does Not Handle

This is not a list of weaknesses. These are deliberate scope decisions that allow DATP-CP to maintain a clean causal isolation and a defensible, pre-registered claim. Disciplined scope control is a methodological strength.

Out-of-scope item	Why excluded from DATP-CP	Where handled	Implication
Training-data poisoning	Different attack surface (training channel). Including it would conflate two mechanisms.	Prior FL-IDS literature	DATP-CP freezes training data
Model / gradient poisoning	Acts on weight channel, not calibration channel. Already covered by Byzantine FL literature.	Byzantine-robust FL literature	DATP-CP freezes model weights
Aggregation poisoning / Byzantine defenses	Aggregation-phase concern. DATP-CP's attack begins after aggregation completes.	FLTrust, Krum, Trimmed-Mean literature	Aggregation untouched
Backdoor / evasion attacks	Different threat class; DATP-CP does not plant triggers or craft adversarial inputs.	Backdoor / adversarial ML literature	Out of scope, future work
Raw-traffic injection realism	Score-level attack does not prove that traffic with target scores can be generated.	Future work (explicitly disclosed)	Score-level proxy limitation stated
Privacy guarantees / formal DP	Privacy is not the research question; data locality is a structural property, not a guarantee.	DP-FL literature	No privacy claims made
Deployment: latency, memory, energy	System-level validation is a separate question requiring hardware evaluation.	Journal extension (system evaluation)	DATP-CP is mechanism-only
Edge-IIoTset / journal-only datasets	Journal extension uses these for generalization. DATP-CP scope is N-BaloT physical devices.	Journal extension (Branch 3)	N-BaloT is primary and sufficient
Model-personalization comparators (FedProx, Ditto)	These modify training and model objectives — incompatible with fixed-model isolation design.	Journal extension (Branch 3)	DATP-CP holds model fixed
Conformal thresholding / temporal drift	Journal-only threshold variants that extend DATP's calibration mechanism study.	Journal extension (Branch 3)	Outside DATP-CP identity

Most objections are answered in context throughout this dossier. This page consolidates the strongest remaining challenges.

Q Why this topic now, after DATP?

A DATP showed that threshold calibration and personalization shape detection behavior. The next logical step is to ask whether that same calibration stage can be adversarially manipulated — an adversarial continuation that follows naturally from the performance study.

Q Why is this not just DATP repeated?

A DATP varies the threshold policy to study performance. DATP-CP fixes all policies and varies only the calibration contamination to study vulnerability. The experimental design, research question, and claim type are entirely different.

Q Why not wait for the journal extension?

A The journal extension strengthens DATP through external validation and comparators. DATP-CP is a separate adversarial branch that does not conflict with or depend on the journal extension's timeline. They can proceed on independent tracks.

Q Why not use more datasets?

A N-BalIoT provides physical-device clients — the specific property needed for device-level calibration behavior. Multi-dataset generalization is a journal-extension goal. DATP-CP's claim scope is explicitly bounded to the nine-device N-BalIoT setting.

Q Why not implement a defense?

A A defense is optional (TRIMMED_CALIBRATION) and enters main results only if it meets pre-specified recovery and regression gates. DATP-CP's primary contribution is the vulnerability characterization, not the defense. Rushed defense design without proper evaluation gates would weaken, not strengthen, the paper.

Q Why is single-client compromise enough?

A The main claim is bounded to single-client compromise. Even one compromised client can shift Global threshold fleet-wide (diluted), shift Local threshold maximally for that victim (undiluted), or create co-cluster propagation under Cluster policy. Single-client is sufficient to isolate the mechanism and study policy differentiation.

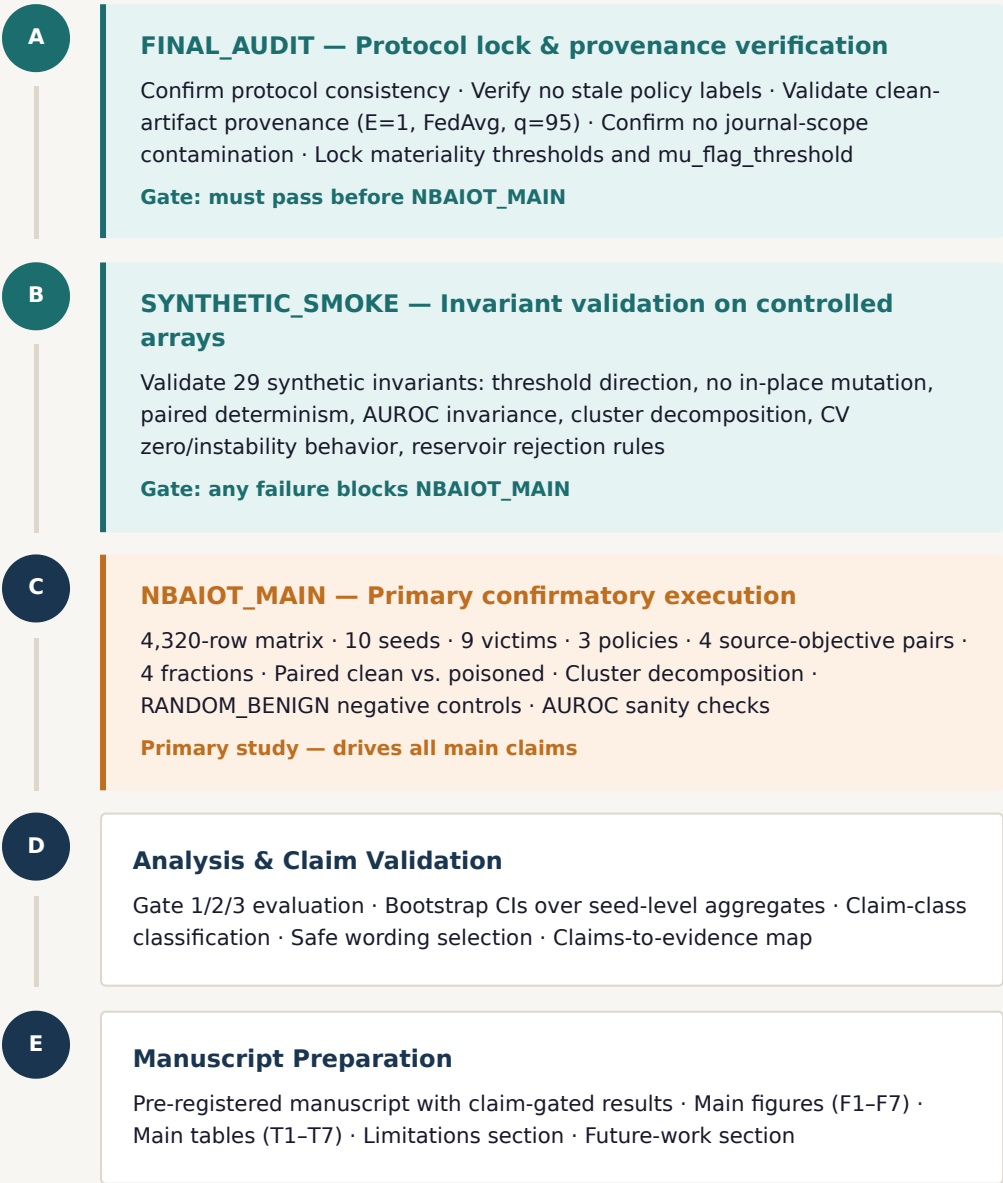
Q Why score-level and not raw traffic?

A Score-level attack cleanly isolates the calibration mechanism. Raw-traffic realizability is a separate question that requires traffic generation capability — an overclaim in this setting. The score-level design is more honest: it shows what the mechanism can do under a controlled model, without implying operational deployment.

Q What is the concrete added value in one sentence?

A DATP-CP isolates a narrow but previously unanalyzed attack surface — the benign calibration buffer — and shows that its contamination can shift detection thresholds and harm detection quality, while leaving models, training, aggregation, and test data entirely unchanged.

Core execution stages



Optional extensions (not required for main claim)

Extension	Condition
Multi-client (pairs + triples)	NBAIOT_FULL_OPTIONAL — feasible compute + pre-registered plan
Optional fraction f=0.05	Locked before execution only
TRIMMED_CALIBRATION defense	Only if Recovery ≥ 0.5 and no clean-regression
Stretch contrast (CICIoT2023)	Diagnostic pseudo-client contrast only; no natural-device claims

Cut order under time pressure

Cut first	Never cut
Stretch contrast	FINAL_AUDIT · SYNTHETIC_SMOKE · NBAIOT_MAIN
Optional defense	Paired clean-vs-poisoned design
Multi-client full extension	All 3 threshold policies
Extra diagnostic variants	Valid source-objective pair validation
Additional fractions	q = 95 lock · AUROC sanity check

Protocol status: The protocol is pre-registered and locked. N-BaIoT main execution begins only after FINAL_AUDIT, SYNTHETIC_SMOKE, and CLUSTER_THRESHOLD_REPRODUCIBILITY all pass.

PROTOCOL

Reproducible Experiment Protocol

Pre-registered protocol with locked seeds, fractions, eligibility rules, reservoir definitions, and materiality thresholds. Manifests record all provenance fields.

CORE RESULTS

Paired Clean vs. Poisoned Result Tables

Threshold-shift summaries ($\Delta\tau$ mean, median, IQR, materiality rate, seed-sign count, victim-majority count, bootstrap CIs) across all policies and pairs.

MECHANISM

Threshold Movement Analysis

$\Delta\tau$ vs. fraction curves by policy and objective. Directional excess over RANDOM_BENIGN negative controls. Random-control instability audit.

DIFFERENTIATION

Policy Comparison Figures

Policy-stratified vulnerability profiles: victim sensitivity, blast radius, spillover count, and cluster decomposition (aggregation, churn, normalization gap).

HARM

Failure-Mode Interpretation

Detection degradation under raising (ΔTPR , ΔBA , $\Delta\text{MacroF1}$) and alarm-burden degradation under lowering ($\Delta\text{CV}(\text{FPR})$, $\Delta\text{IQR}(\text{FPR})$, $\Delta\text{WorstClientFPR}$).

MANUSCRIPT

Claim-Bound Paper

Conference paper with claim-gated results, reviewer-risk controls, safe wording, and explicit limitations section. Target: security or networking venue.

Main figures and tables planned

Figure	Content	Table	Content
F1	DATP-CP attack surface — pipeline with calibration highlighted	T1	Protocol identity and threat model
F2	Score-level poisoning mechanism — fixed-size replacement	T2	Experiment matrix
F3	Threshold shift vs. fraction (per policy, per source-objective pair)	T3	N-BaloT client and eligibility summary
F4	Detection degradation under threshold raising (victim ΔTPR)	T4	Main threshold-shift results
F5	Alarm-burden degradation under threshold lowering ($\Delta\text{CV}(\text{FPR})$, ΔIQR)	T5	Downstream movement
F6	Policy vulnerability profile comparison	T6	Policy-differentiated vulnerability profile
F7	Cluster-threshold decomposition (aggregation, churn, normalization)	T7	Claims-to-evidence map
		A1	TRIMMED_CALIBRATION diagnostics (optional)

All figures and tables correspond to pre-registered analysis steps. F1–F3 document the mechanism; F4–F5 document the harm; F6–F7 document policy differentiation. T7 maps all claims to evidence gates.

7 main figures
7 main tables + 1 optional

These limitations are pre-registered and explicitly disclosed in the manuscript. Acknowledging them strengthens, not weakens, the scientific credibility of the study.

Limitation / Risk	Why it matters	How DATP-CP bounds it	Status in the paper
Score-level proxy	DATP-CP operates on reconstruction scores, not raw traffic. The attack may not be realizable by traffic generation.	All raw-traffic realizability claims are explicitly excluded. Score-level is disclosed as a proxy for the calibration mechanism.	Disclosed in abstract, scope section, and limitations
N-BaloT scope (9 devices)	Nine devices may not generalize to larger or more diverse federations.	Claims are bounded to "the tested nine-device N-BaloT setting." Stretch contrast is diagnostic only.	Disclosed as dataset boundary throughout
Single-client compromise only (main)	Multi-client collusion could amplify effects beyond what single-client results show.	Main claims cover single-client only. Multi-client is optional extension with separate pre-specified gates.	Scope stated in abstract and protocol
Lowering is conditional	Random fixed-budget insertion of low scores may fail to displace the p95 threshold's upper-tail support.	Pre-registered as conditional. Diagnostic targeted-removal variant distinguishes robust channel from weak instantiation.	Disclosed in attack design and results
With-replacement score duplication at high f	At $f=0.40$, resampling can produce duplicate score values in the calibration buffer.	Disclosed as proxy limitation in the manuscript. Not hidden as realistic traffic generation.	Disclosed in attack design section
CV(FPR) undefined at zero mean FPR	If mean eligible-client FPR is zero, CV is undefined and cannot support lowering claims.	Absolute dispersion metrics (IQR(FPR), max–min FPR, WorstClientFPR) are always reported as backups. CV cells are excluded from claims when undefined.	Handled in metrics definition and contingency plan
No defense in main (optional only)	A reviewer may request at least a preliminary defense.	TRIMMED_CALIBRATION is included optionally with pre-specified gates. No defense is framed as effective without meeting those gates.	Optional appendix if gates pass
AUROC invariance expected — not a contribution	A reviewer might interpret AUROC invariance as "the attack had no effect."	AUROC invariance is a sanity check that test scores and labels were not changed. The contribution is threshold-induced operating-point movement, which AUROC does not measure.	Explained explicitly in results and discussion

The three-step thesis progression

1

DATP — Threshold personalization matters

A shared global threshold imposes $CV(FPR) = 1.017$ across heterogeneous physical IoT devices. Per-client calibration reduces this to 0.299. The threshold-calibration stage is the decisive variable for detection equity.



2

DATP-CP — Threshold calibration is also vulnerable

Score-level contamination of a single client's calibration buffer shifts detection thresholds without altering training, aggregation, model parameters, or test data. Raising attack: 100% gate pass, -2.4 to -7.9 pp victim TPR. Lowering: $\Delta CV(FPR) +0.05-0.06$ for Local and Cluster. Policy choice governs harm magnitude and propagation.



3

Journal Extension — DATP result survives generalization

External validation on Edge-IIoTset, model-personalization comparators (Ditto, FedProx), temporal recalibration, and additional threshold variants confirm that DATP's threshold-scope result is robust — handled separately, does not absorb DATP-CP's adversarial branch.

Central thesis: what DATP-CP proves

"DATP-CP's added value is not breadth — it is controlled isolation of a previously under-analyzed calibration-channel attack surface in federated IoT anomaly detection."

Three claims the dossier defends

1. The gap is real. Training- and aggregation-phase FL defenses do not observe the post-training threshold-calibration stage. This structural gap exists regardless of whether a specific attack is practical.

2. The gap is exploitable at the score level. Under a gray-box score-access threat model and controlled score-level contamination, the calibration channel can be shifted materially enough to cause downstream detection harm.

3. Policy choice matters for both vulnerability and propagation. Local, Global, and Cluster threshold policies differ in victim sensitivity, blast radius, and spillover — the same contamination yields qualitatively different outcomes under different policies.

DATP-CP · Calibration-Channel Poisoning Study

LAMSAD Laboratory · Hassan First University · 2026

Salaheddine Naslouby · Mohamed Lachgar · Hamid Hrimech